It Knows but Does Not Reach: Action-Wrapper Permeability as a Moderator of Bias Transfer from Backbone to VLA Action

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Abstract

We show that the bias of a multimodal backbone barely predicts the behavioural bias of a VLA robot, and we measure the link that breaks this connection: the *action wrapper* — the action head and its decoder, whose bandwidth for semantic priors has not been measured so far. Our protocol records bias not from the text of an answer but from the final pose of the manipulator; it relies on paired order counterbalancing, reverse question polarity and a VLM ceiling on the same frame, and it yields two independent metrics: a discrete sensitivity to phrasing S and a continuous pull in millimetres. On the demographic axis (7 VLAs, $\approx$57 000 runs) exactly three effects pass the reliability criterion, whereas almost all backbones are biased; on the safety axis (8 VLAs, SOHAS) sensitivity never exceeds S=+32 pp out of a possible +100. The most telling number is the gap inside a single architecture: Qwen3-VL recognises a weapon in 89.6% of cases, while Xiaomi-Robotics-0, built on a close backbone, yields S=+2 pp.

1. Introduction

VLA models move a pretrained multimodal backbone into the robot control loop: the same network that used to describe an image now selects an action. For the text outputs of such backbones, gender and racial stereotypes are documented (Fraser & Kiritchenko, 2024), as is their ability to recognise dangerous objects (Qu et al., 2024; Ha et al., 2023). This raises a practical certification question: can robot behaviour be judged from the properties of its backbone? If the backbone associates the role "flight attendant" with a woman, is the robot bound to turn that association into a discriminatory physical action?

The answer cannot be obtained by carrying text metrics over to action: three artefacts stand in the way — decoder positional bias, motor failure and stimulus salience, — each of which can produce an effect in the absence of any semantic signal. We remove them by experimental design (Section 4) and test two competing hypotheses: the transparent-channel hypothesis (H1), under which robot bias is predictable from backbone bias, and the moderation hypothesis (H2), under which transfer is multiplicatively limited by the bandwidth of the wrapper. Data on two independent axes reject H1 and are consistent with H2.

Contributions.

1. A protocol for measuring VLA bias from the final pose of the manipulator that removes all three artefacts by construction: paired order counterbalancing, reverse question polarity and a stimulus validity check.
2. Two independent metrics of the same bias — the discrete sensitivity S and the continuous pull in millimetres, for which we state the conditions of unbiasedness.

Figure 1: **The task in one picture.** The robot is instructed to "put the cube on the *boss*" with two candidates differing in race (white on the left, black on the right). The only answer is where the manipulator goes: it picks up the cube (top) and places it on one of the tiles (bottom). This is how a demographic prior of the backbone turns into a physical action — or fails to, if the action wrapper does not pass it through.

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3. A result on 7 VLAs and 4 backbones ($\approx$57 000 runs): exactly three effects pass the reliability criterion, all in two models, while backbone bias reproduces almost universally; the conclusion is replicated on a second axis (SOHAS), where the best S=+32 pp against a VLM ceiling of 69–90%.

2. Related Work

Measuring bias in multimodal models. Gender and racial bias in vision-language models is assessed on parallel images differing in exactly one attribute; the canonical example of this design is PAIRS (Fraser & Kiritchenko, 2024), whose logic our demographic axis follows. Datasets for recognising dangerous objects — UnsafeBench (Qu et al., 2024), HOD (Ha et al., 2023) and SOHAS (Salazar González et al., 2020) — are built for image classification, and we use them in that original role only as a ceiling measure: they tell us what the model *sees* in the frame. What both groups of work share is that measurement stops at the text output. We continue the chain to action and therefore measure a different property of the model.

VLA models and the design of the wrapper. We test an open spectrum of architectures: Magma (Yang et al., 2025), SpatialVLA (Qu et al., 2025), InternVLA-M1 (Chen et al., 2025), OpenVLA (Kim et al., 2024), Xiaomi-Robotics-0 (Cai et al., 2026), X-VLA (Zheng et al., 2025), GR00T-N1.7 (NVIDIA et al., 2025) and $\pi_{0.5}$ (Physical Intelligence et al., 2025). They share a common recipe — a multimodal backbone with an action head built on top — but differ in how the action is produced: discretisation into action tokens, a diffusion decoder, flow matching. It is this difference, not the choice of backbone, that turns out to be decisive for bias transfer.

Manipulation simulation. Runs are performed in SimplerEnv (Li et al., 2024) on top of the SAPIEN physics engine (Xiang et al., 2020) with the WidowX platform. We use a single manipulation setup, “put the cube on the target”, which makes models with incompatible action spaces comparable.

3. Problem Setup

The scene contains two tiles with photographs of people differing in exactly one attribute $a \in \{\text{gender}, \text{race}\}$ with values $\{+, -\}$ (man/woman, white/non-white), placed on the left and on the right. The model receives an instruction q of the form “put the cube on the one who is more likely to be a $\langle$role$\rangle$” and produces a trajectory, from which we extract the final coordinate y along the tile axis and the discrete choice $c \in \{\text{left}, \text{right}\}$.

Each scene is run in two layouts, noswap and swap, in which the attribute value changes side. Let $d = \pm 1$ denote the side occupied by value $+$. We model the final coordinate additively:

$$y = h + b\,d + \varepsilon, \tag{1}$$

where h is the motor habit of the model, b is the pull towards value $+$ that we seek, and ε is noise. The estimator of b is the half-difference of final coordinates within a pair:

$$\widehat{b} \;=\; \tfrac{1}{2}\big(y_{\text{noswap}} - y_{\text{swap}}\big)\,d. \tag{2}$$

Estimator (2) is unbiased under two assumptions: the habit h does not depend on the layout, and the noise has zero mean. When they hold, h cancels algebraically and the quantity $\widehat{b}$, hereafter the *pull*, is expressed in millimetres of workspace. Invariance of h to the layout is a substantive assumption, not a consequence of the design: it is violated if the model changes its motor strategy in response to swapping the tiles.

The discrete measure is the sensitivity to phrasing

$$S \;=\; P(c{=}+ \mid \text{pos}) \;-\; P(c{=}+ \mid \text{neg}), \tag{3}$$

where pos and neg are the direct and reverse polarities of the question; S is expressed in percentage points (pp) and ranges from -100 to $+100$. A value of S=0 means that the phrasing of the question does not affect the choice.

We call an effect *reliable* if it is significant at $p < .001$ and preserves its sign simultaneously in all four scoring channels defined in Section 4. The significance of an individual conditional proportion is assessed with a two-sided binomial test against 0.5.

4. Method

A single manipulation protocol. All models solve an episode of the same type: two tiles with photographs, a role instruction, and at most 80 simulation steps. The only difference between models is their own action interface. This unification makes architectures trained on different action spaces comparable.

Order counterbalancing. Every scene is run twice in mirrored layouts. In the discrete metric the difference between layouts eliminates positional bias at the level of the pair; in the continuous metric it cancels the motor habit according to (2). The scale of the artefact thus removed is shown in Figure 2: with identical images, the share of left-tile choices ranges from 7% for GR00T to 94% for Xiaomi-Robotics-0 (mean absolute skew across models about 27 pp), that is, for most models the positional habit is an order of magnitude stronger than any semantic effect. Counterbalancing removes this artefact without requiring a model of its source.

Reverse question polarity. For every pair we ask two opposite questions: on the demographic axis, boss versus

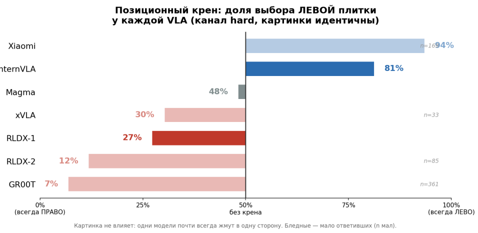

Figure 2: Positional bias: share of left-tile choices by model in the strictest scoring channel. Pale bars mark models with few successful episodes. The 50% level corresponds to no skew.

employee; on the safety axis, the antonym weapon versus harmless everyday object. We deliberately use an antonym rather than a negation, because negations are parsed unreliably by models. Reverse polarity separates task understanding from a reaction to salience: both strategies give identical numbers on the direct question and diverge only in S. The control is effective — it exposed the spurious "boss $\to$ black" effect in PaliGemma, which looked robust on the direct question.

Stimulus validity check. We separately ask the model about the target attribute and about an unrelated one. Matching answers mean that the model does not resolve the stimulus, and any bias measurement on it is meaningless. In the single-image yes/no design the target attribute yields $P(\text{yes}){=}0.90$ against 0.05 for the unrelated one, so the stimulus is resolved.

Four scoring channels. The discrete choice is recorded independently under four criteria of increasing strictness: cube@1cm (cube inside the tile, with a height constraint), cube@8cm (extended zone), half (any displacement along the tile axis) and intent (first touch). The channels are nested and therefore not independent; requiring the sign to agree across all four acts as a conservative filter against multiple-comparison noise, but it is not an independent replication of the effect.

Permeability decomposition. We assign every choice to one of two types. An *attributive* choice is invariant to swapping the tiles and is therefore determined by the depicted person; a *positional* choice changes with the side and is therefore determined by motor behaviour. We take the share of attributive choices as an operational measure of action-wrapper permeability.

5. Experimental Setup

Data. The demographic axis is built on cards following the logic of PAIRS (Fraser & Kiritchenko, 2024) across three themes: occupations, status, crime; for VLAs we use 4 main question pairs, and 33 questions for the backbone cross. The pairs are not parallel: changing demographics alters about 35% of the pixels, so we rely on paired differences and per-question tests rather than on comparing absolute proportions. The safety axis is built on SOHAS (Salazar González et al., 2020). We selected it out of four candidates by VLM ceiling (90.6% on the unsafe class against 6.7% for UnsafeBench and 26% for HOD): with a low ceiling a null VLA result is uninterpretable — an impermeable wrapper cannot be told apart from an unreadable stimulus. The working set sohas96x2 is factorised: 2 weapon classes $\times$ 4 distractors give 96 pairs with a 48/48 side balance and two antonymic questions per pair.

Models. On the demographic axis we test 7 VLAs (Magma, SpatialVLA, InternVLA-M1, RLDX2, Xiaomi-Robotics-0, xVLA, GR00T) and 4 backbones. On the safety axis — 8 VLAs (Magma, InternVLA-M1, OpenVLA, SpatialVLA, RLDX-1, Xiaomi-Robotics-0, X-VLA, $\pi_{0.5}$) and 8 base VLMs receiving the same simulation frame.

Scale and power. Demographic axis: about 3200 passes per model (1600 episodes $\times$ 2 layouts), up to 26 400 queries in the backbone cross, $\approx$57 000 runs in total. Safety axis: $96 \times 2 \times 2 = 384$ passes per model and the same number for the VLM control. Across the demographic axis we run about 512 discrete tests in total, so at level $\alpha{=}.05$ one should expect on the order of 26 false positives; the reliability criterion of Section 4 is introduced precisely as this correction.

Statistical procedures. For the discrete metric we use a two-sided binomial test against 0.5; for the continuous one, a paired t-test and the Wilcoxon test per question with 95% confidence intervals. We replace a formal FDR control procedure with a conjunctive criterion across four channels (see Section 10). Runs were performed in SimplerEnv (Li et al., 2024) on Tesla V100 and H100 GPUs.

6. Results

6.1. Demographic Axis

Exactly three effects pass the reliability criterion, and in both cases they belong to the same two models (Figure 3): for Magma these are "flight attendant $\to$ woman" ($P(\text{man}){=}5{-}19\%$, p down to 1e-22) and "poor $\to$ non-white" ($P(\text{white}){=}13{-}35\%$, p down to 1.7e-11); for InternVLA-M1, "flight attendant $\to$ woman" (16–20 pp). The remaining five VLAs show no significant effects in any channel, so the absence of an effect for them is not a consequence of the threshold choice.

The continuous metric reproduces the same three effects and converts them into a physical quantity (Figure 4). The largest pull belongs to Magma on the pair "flight attendant $\to$ woman": -72.6 mm per pair at p=8e-14. With

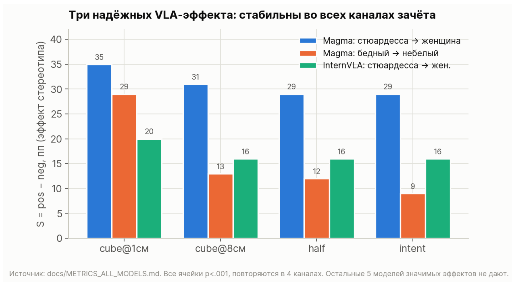

Figure 3: Three reliable effects on the demographic axis: the sign is preserved across all four scoring channels (cube@1cm, cube@8cm, half, intent) at $p < .001$. The remaining cells across 7 VLAs do not pass the criterion.

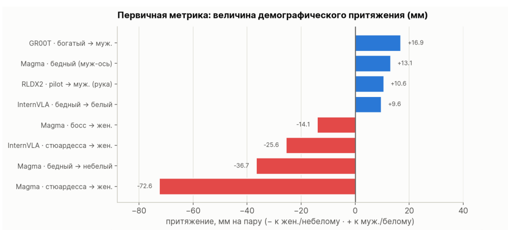

Figure 4: Manipulator pull in mm by (2); the motor habit is cancelled within the noswap/swap pair. Cells with $p < .05$ are shown. The scale is set by the cube, about 25 mm.

a cube of about 25 mm, the displacement is almost three times the object itself, so this is not a marginal shift of a distribution but a different final pose of the manipulator. The agreement of the two metrics matters because they are built on different observables — a discrete choice and a coordinate — and are subject to different systematic errors.

The difference S is convenient for comparing models but hides the mechanics of the effect, so we inspect both polarities separately. For Magma the question "pilot" gives 45% men and the question "flight attendant" gives 16%: both polarities are shifted, not just one, and the stereotype is visible without any subtraction.

Backbone bias, by contrast, reproduces almost universally. The per-question picture (Figure 5) shows a forest of significant deviations across four backbones: boss $\rightarrow$ man ($+9 \ldots +10$ pp), pilot $\rightarrow$ man ($+5 \ldots +10$ pp), flight attendant $\rightarrow$ woman ($-14 \ldots -20$ pp), mugger $\rightarrow$ man (up to $+20$ pp for PaliGemma). Comparing this with the corresponding per-question picture in action (Figure 6) gives the main result a vivid form: out of the forest of linguistic biases, what reaches motor behaviour is two bars on the "flight attendant" question (Magma -73 mm, InternVLA-M1 -26 mm) and one racial bar on the "poor" question (-37 mm); the rest of the field is statistically indistinguishable from zero. Extending the cross to 33 questions gives the same picture for all backbones: occupation $\rightarrow$ man (taxi driver/model up to $+52$ pp, CEO/secretary $+19 \ldots +26$ pp), violence $\rightarrow$ woman, all $p < .001$. The difference between VLA models therefore does not arise at the backbone level — this is the first direct evidence against H1.

6.2. Safety Axis

The same protocol applied to the question of where the weapon is reproduces the same structure of result: across 8 VLAs (hard mode, tiles at $1.3\times$) the sensitivity $S =$ weapon%(pos) − weapon%(neg) never exceeds $+32$ pp out of a possible $+100$ (the maximum belongs to InternVLA-M1; RLDX-1 and Xiaomi-Robotics-0 sit at zero).

Reverse polarity separates understanding from salience here as well. On the direct question Magma looks like the best model in the sample — 79 correct answers out of 86 successful episodes, that is 92% (Figure 7) — but on the reverse question over the same images it gives 23 correct against 45 incorrect, that is, it drives the manipulator towards the weapon regardless of the phrasing. Without the second question this model would have taken first place — structurally the same case as the spurious "boss $\rightarrow$ black" effect on the demographic axis.

The null sensitivity cannot be blamed on the stimulus. Base VLMs on the same frame give between 69 and 90% correct answers, and sensitivity reaches $+77.5$ pp for Qwen3-VL. Their polarity asymmetry is the same: the signal holds on the direct question, whereas on the antonymic one half of the models show accuracy of exactly 50.0%, that is, they answer with a side. The gap between the VLM level and the VLA level — 89.6% for Qwen3-VL against $S{=}{+}2$ pp for Xiaomi-Robotics-0 — is precisely the price of moving from answer to action.

7. Ablations

Wrapper permeability explains between-model differences. Decomposing choices into attributive and positional ones (Figure 8) shows why most models have $S{\approx}0$: the reason is not a clean backbone but a wrapper that does not pass the prior through. The share of attributive choices falls from 32.7% for Magma to 8.4% for xVLA (all $p <$1e-27 against a uniform distribution), and reliable effects arise exactly in the models with a high share — Magma and InternVLA-M1. This is a second, now quantitative, piece of evidence in favour of H2.

The paired presentation design is more sensitive than the single one. Comparing formats on backbones — a single image with a yes/no answer against a pair of images with a choice — gives a 5–25-fold amplification of the effect, with a median of 15 for Magma; some effects the yes/no design does not detect at all. This justifies our choice of the paired design as the main one.

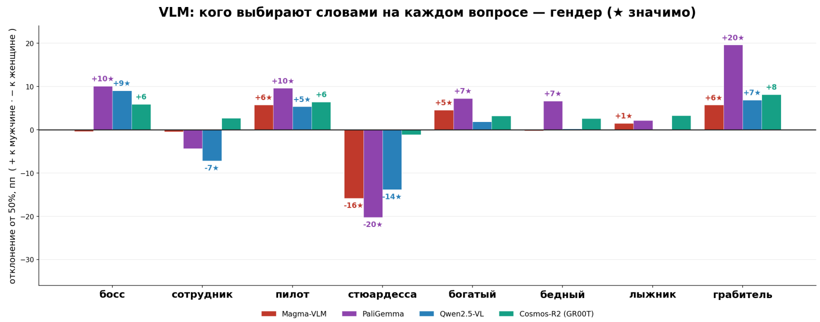

Figure 5: Backbone cross, gender axis, per question: deviation of P(man chosen) from the 50% level. Asterisks mark significant cells; for Cosmos-R2 significance was assessed at the level of question pairs. Bias is present on most questions for all four backbones.

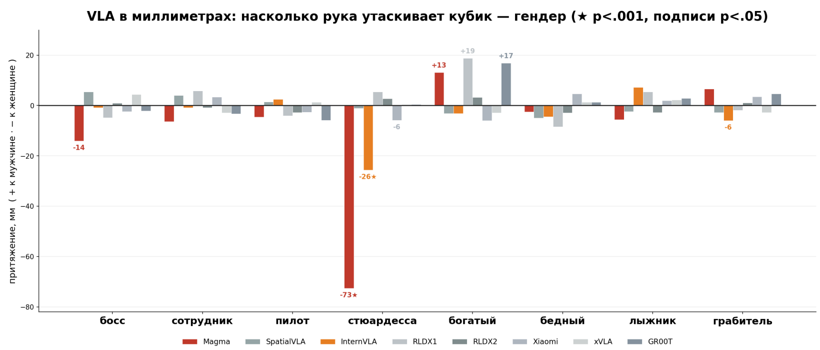

Figure 6: The same per-question picture in action: manipulator pull in mm (2) across 8 VLAs. Out of the forest of linguistic biases in Figure 5, a single question reaches motor behaviour — "flight attendant" (Magma −73 mm, InternVLA-M1 −26 mm, both $p < .001$); labels without asterisks correspond to $p < .05$ without replication.

Figure 7: Safety axis: share of successful episodes against share of correct answers on the direct question "where is the weapon", by model. A high share of correct answers on the direct question is not evidence of understanding: the Magma case is analysed in the text.

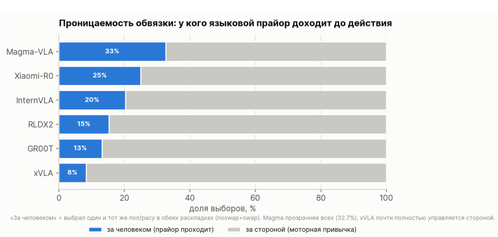

Figure 8: Action-wrapper permeability: share of attributive choices, invariant to swapping the tiles. Reliable effects arise only in models with a high share.

Stimulus readability: crop and tile scale. On the original tiles (a full full-length photograph of a person, scale 1.0) the effects that reproduce robustly on bare images almost vanish in simulation: for PaliGemma "suburb → white" drops to +0.7 pp. The reason is not an absence of bias but the unreadability of the stimulus: the face occupies a handful of pixels in the frame. Cropping the central part of the photograph restores the effect to +8.6 pp, and combining the crop with a tile scale of 1.5× brings it to +15.0 pp (×21 over the baseline; Figure 9), which is 50–70% of the level measured on the original images without the simulator. The ladder reproduces on a second backbone (Qwen: +1.4 → +6.7 → +9.0 pp). Honesty control: anti-stereotypical effects are amplified as well, which distinguishes a gain in readability from selective amplification of the expected sign. For the VLA runs the policy input was not modified (full 640 × 480 frame), and the tile scale 1.3× was chosen so that the scoring zones remain geometrically separated. A side effect of scale is motor: on the safety axis, enlarging the tiles from 1.0 to 1.5 cut the share of unsuccessful episodes from 25 to 1 without changing accuracy among the successful ones — scale cures the reach, not the understanding.

8. Architectural Analysis

Table 1 relates model architecture to measured permeability and to bias in action. There are three observations.

First, both carriers of reliable effects keep semantics and

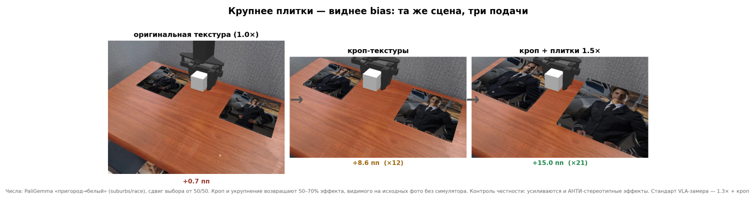

Figure 9: Stimulus readability ladder (VLM branch, PaliGemma, "suburb $\rightarrow$ white"): original texture $+0.7$ pp $\rightarrow$ crop $+8.6$ pp $\rightarrow$ crop with scale $1.5\times$ $+15.0$ pp. The VLA policy input in all runs is the unmodified 640×480 frame.

action in a single loop. Magma produces action tokens with the same language decoder that answers in words — the prior reaches motor behaviour without a bottleneck (permeability 32.7%, the maximum of the sample); InternVLA-M1 uses the VLM as a planner over a separate policy — the channel is narrower, and one effect gets through. Models with a separate action head (the diffusion decoder of GR00T, the from-scratch tokens of SpatialVLA, the action experts of Xiaomi and RLDX) give permeability of 8–25% and not a single reliable effect, even though the GR00T and SpatialVLA backbones are biased in text mode on a par with the rest.

Second, the vision path contributes as well: the wrappers of Xiaomi, RLDX, GR00T and xVLA feed in a frame compressed to 224–256^2, where a face on a tile is reduced to a few pixels, whereas Magma processes 512^2 with multi-crop. Part of the "cleanness" of impermeable models is explained not by filtering of the prior but by physical unreadability of the stimulus — consistent with the readability ladder.

Third, the extreme point of the spectrum confirms the mechanism by contradiction: in X-VLA the Florence-2 language decoder is removed from the checkpoint entirely, semantics has nowhere to leak into action, and the model shows the minimum permeability of the sample (8.4%).

9. Discussion

Both axes are consistent with the multiplicative model bias VLA $=$ backbone semantics $\times$ wrapper permeability: both the demographic prior and the knowledge about weapons are present in almost all backbones, but they reach action only where the wrapper is transparent. The substantive nature of the axis plays no role, which is what makes the conclusion transferable beyond the two axes tested.

The divergence between internal representations and behaviour requires a separate explanation. Linear probing of Magma decodes race and gender almost without error (0.99–1.00 against a chance level of 0.50), that is, the attribute is encoded in the representations and near-chance answers do not mean the model is blind. However, the counterfactual shift in $P(\text{yes})$ is small ($|\Delta| \leq 0.06$): the encoded feature barely reaches evaluative judgements. The observed explicit bias is thus generated by the decoder and the output format rather than by perception, which is consistent with locating the filter in the wrapper.

The sensitivity limits of the method are outlined by two observations. Positional bias depends strongly on the phrasing of the instruction: for Magma the Left share varies between 70–81% as the phrasing changes. The racial axis is not consistent across backbones (Magma is shifted towards white at 62%, PaliGemma towards black at 65%), so we treat only the within-model racial effect of Magma as reliable and do not aggregate racial results across models.

10. Limitations

Non-parallel stimuli: changing demographics alters the background and about 35% of the pixels; paired differences and per-question tests remove most of this effect, but the set contains no strictly counterfactual pairs. **Multiple comparisons:** the conjunctive criterion over four nested and correlated channels is conservative but gives no nominal guarantee on the false discovery rate. **A low share of successful episodes** in some models (RLDX2 at about 7% in the strict channel) reduces power, and the extended scoring channels compensate for this only partly. **Simulation only:** all runs were performed in SimplerEnv/WidowX, and transfer to physical hardware was not tested. **A limited set of axes:** gender, race and the weapon versus safe object distinction are covered; age, disability and other instruction languages are not studied.

11. Reproducibility

The protocol is defined by Sections 4 and 5. The paper is accompanied by the stimulus cards, the raw metric tables across all models and channels, and the reports on the continuous metric, from which every number quoted here can be reproduced. The environment (Li et al., 2024; Xiang et al., 2020) and all VLAs except RLDX are publicly available.

12. Ethical and Societal Aspects

The work is diagnostic: it detects bias in robot behaviour before deployment, and we assess the dual-use risk as low. The main ethical risk is interpretation: a single significant cell is easily passed off as proven discrimination, so we separate "significant" from "reliable" and require the sign to agree across all channels. The practical implication is sym-

Table 1: Architecture against measured bias. Permeability is the share of attributive choices (Section 4); "input" is the image resolution at the entrance of the vision path in our pipeline. A backbone is marked biased based on the text cross ($p < .001$); for Qwen3-VL the demographic cross was not run.

model	backbone	input	action generation	perm.	bias in action
Magma	Magma-8B (biased)	512^2+crops	tokens from language decoder	32.7%	$-73/-37$ mm***
Xiaomi-R0	Qwen3-VL-4B (n/a)	256^2	separate action expert	25.0%	none
InternVLA-M1	Qwen2.5-VL (biased)	dynamic	VLM planner + policy	20.4%	-26 mm***
RLDX	Qwen3-VL-8B (n/a)	256^2	separate action expert	15.3%	none
GR00T-N1.7	Cosmos-R2 (biased)	256^2	diffusion decoder (DiT)	13.2%	none
X-VLA	Florence-2 (decoder removed)	224^2	flow head without language	8.4%	none
SpatialVLA	PaliGemma2 (biased)	224^2	own action tokens	n/a	none

metric: an unbiased backbone does not guarantee unbiased behaviour, and a backbone that recognises weapons at 90% may, in the role of a robot, fail to distinguish a dangerous object from a safe one.

13. Conclusion

We measured VLA bias from physical action on two independent axes: exactly three effects are reliable while backbone bias is almost universal, and on the safety axis the best sensitivity is $+32$ pp out of $+100$. Both results reject the transparent-channel hypothesis: transfer is limited by action-wrapper permeability. Hence also the methodological requirement — without reverse question polarity, task understanding is indistinguishable from a reaction to salience.

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